Week 4 — Milestone 2: Project Plan v1

Journey Maps

Each map follows one persona through Masari. The principle is User Control and Agency. Features named below come from the Week 2 selected concept unless marked proposed.

Journey Map 1: Omar Salah (student)

Stage	Persona actions and feelings	Pain points	Where User Control and Agency helps
1. Start	Opens Masari on the family tablet. Reads the first screen. Feels nervous and unsure what to press.	Long first screen. Fear of pressing the wrong thing. Wi-Fi may drop.	Start screen is short, says Masari is an AI guide, and shows that he can skip, go back, or leave at any time. He sets the pace.
2. Ask	Looks at pathway cards. Answers the interest questions one by one. Types a short question when a word is unclear.	Many questions in a row. Unfamiliar terms.	He can skip any question and come back later. Questions appear in small steps. A plain-language option is available.
3. Understand the answer	Reads a short list of pathways worth exploring. Feels unsure whether Masari is telling him what to do.	Dense explanations. May read a suggestion as an order.	Suggestions are presented as options to explore, not decisions. Reasons stay short and he opens more detail only if he wants it.
4. Act or fix a problem	Notices he answered one question by mistake. Changes the answer and watches the suggestions update. Asks something Masari cannot answer.	Not knowing how to go back. Fear that the change cannot be reversed.	He can edit any answer and suggestions update. A clear message says when information is unverified or out of scope.
5. Finish	Writes a short note about what he liked. Shows it to his parents. Leaves without choosing yet.	Feeling pressure to decide now.	The final decision stays with him. He can leave at any time and use notes to discuss with his family.

Journey Map 2: Hoda Ramadan (parent)

Stage	Persona actions and feelings	Pain points	Where User Control and Agency helps
1. Start	Opens Masari on her smartphone. Wonders what an AI chatbot is and what happens to what she types. Reads what Masari is and is not.	Unsure what the AI is, what it knows, and what it does with her information. Weak signal at home.	Start screen says it is an AI guide, what verified information it uses, that it never asks for personal data, and that suggestions are for exploring, not deciding. The final decision stays with the family. She chooses where to begin.
2. Ask	Looks at the pathway cards. Asks the guide in Arabic, “What is the difference between these two pathways?” Turns on simple wording when a term is new.	New education terms and rules. Different opinions from neighbors and relatives. Fear of pressing the wrong thing.	Pathways are compared on the same criteria, with none labeled better. The guide answers only from verified information. The “Simple wording” switch gives plain Arabic. If Masari is not sure, it says so and points to the school counselor or the Ministry.
3. Understand the answer	Reads short answers with their reasons and the source named. Compares them with what relatives told her. Wonders whether the tool is pushing her daughter toward one pathway.	Cannot tell official information from rumor. Worry that the tool will steer her daughter. Needs a source she can name when she asks others.	Each answer cites its source and shows reasons and uncertainty in plain language. Unverified details are flagged. Suggestions are options to explore, not decisions, and no pathway is called the best.
4. Act or fix a problem	Finds an answer or a flagged detail she does not understand. Switches to simple wording and asks again. Checks the cited source. The connection drops once.	Hard to tell whether a flagged detail is wrong or just unconfirmed. Fear of losing what she read when the signal drops.	Unverified details are flagged, not hidden. She can ask again in simpler words without starting over. A fallback message appears on errors and time-outs.

Stage	Persona actions and feelings	Pain points	Where User Control and Agency helps
5. Finish	Writes short notes on what she understood and the questions she has for the school counselor. Rereads them later. Talks with her daughter.	Pressure to choose for her daughter. Remembering what she read.	Notes keep short points she can reread later and take to the counselor. The final decision stays with the family.

Journey Map 3: Hassan Nour (school counselor)

Stage	Persona actions and feelings	Pain points	Where User Control and Agency helps
1. Start	Opens Masari on his laptop in the school office. Looks for the model, the data, and the limits before using it with students and parents.	Chatbots that do not say what they use or cannot do. Little time for each check.	Start screen discloses that it is an AI, what verified information it uses, that rankings come from documented rules and not the LLM, and that the final decision stays with the student and family.
2. Ask	Tries Masari with a fictional profile, never a real student. Asks detailed follow-up questions in Arabic about the rules and their sources.	Shallow answers without sources. Confident answers he cannot check.	Answers draw on verified information only and cite their source IDs. When evidence is weak, Masari says so instead of guessing.
3. Understand the answer	Reads the factors behind each suggestion and checks them against the official decree text. Worries that students will read a result as a decision.	Scores that look too precise. Cannot see how a suggestion was produced. Rules change and official texts are hard to find.	Reasons show the factors actually used, and no score is presented as a prediction of ability or success. Each pathway record stores its article, source link, and date checked. Showing them to users is proposed.

Stage	Persona actions and feelings	Pain points	Where User Control and Agency helps
4. Act or fix a problem	Adjusts answers and factor importance and watches how the suggestions change. Finds a fact that looks unverified or does not match the official rules, and reports it.	Tools that cannot be tuned. No way to report that something looks wrong.	He edits inputs and factor weights and sees the effect. Unverified details are flagged. Reporting a bad suggestion with an optional reason (thumbs-down) is .proposed
5. Finish	Saves his notes and the alternatives he compared. Uses them to prepare his guidance sessions with students and parents.	Little time per student. Losing his comparison work.	Notes record his own reasoning and support his sessions, not replace them. The final decision stays with the student and family.

Measurable Objectives

These targets are the goals adopted for Plan v1. They are goals, not results. They will be measured in Weeks 7 to 10 and reported honestly, including any that are missed.

AI-quality test set: 30 fixed, labeled prompts, written in Week 5: 10 normal, 8 edge cases, 6 out-of-scope, and 6 risky. The same set is used before and after each improvement.

User tests: At least 5 adult testers from outside the team will participate, with informed consent and anonymous codes (T1 to T5). Testers will role-play the three primary user personas: a Grade 9 student, a parent, and a school counselor. No minors are involved in this project. This is a limitation: the results will reflect adult testers' role-play experiences rather than the actual responses and behaviors of real Grade 9 students, parents, or school counselors.

Note on sample size: With 5 testers, a percentage target corresponds to whole testers (for example, 4 of 5 = 80%, so a 90% target requires 5 of 5). The SUS mean from 5 testers is still indicative, not statistically meaningful. The team decided to use 5 testers (T1 to T5).

AI-quality metrics

Metric	How it is measured	Target (proposed)	How data is collected
AQ1. Groundedness	Share of factual claims in answers that are supported by the verified sources.	≥ 90% of claims supported	The evaluation script runs the test set and saves answers. Two team members label each claim as supported or unsupported in a spreadsheet and settle disagreements together.
AQ2. Safe and out-of-scope handling	Share of out-of-scope or risky prompts handled correctly (for example, asking Masari to pick for the student, predict success, or answer unrelated topics).	≥ 90% handled correctly	The 12 prompts labeled out-of-scope or risky. Two team members mark each response correct or incorrect.
AQ3. Edit-response accuracy	Share of student edits (changed answer, skipped question, changed factor importance) that change the suggestions exactly as the documented scoring rules predict.	100% of tested edits	Scripted edit cases on fictional profiles. The script compares output with the expected result from the rules.
AQ4. Arabic answer quality (proposed)	Share of Arabic answers on the 30-prompt test set rated clear and correct by two team members.	≥ 85% of Arabic answers rated clear and correct	The evaluation script runs the 30-prompt test set and saves the Arabic answers. Two team members rate each answer as clear and correct or not, and settle disagreements together.

User-experience metrics

Metric	How it is measured	Target	How data is collected
UX1. Task success	Share of observed tasks finished without help.	≥ 80% “yes”	Observed usability test with 3 to 5 written tasks per tester. Record yes, partly, or no for each task.

Metric	How it is measured	Target	How data is collected
UX2. Usability (SUS)	Average System Usability Scale score.	Mean ≥ 68	Standard 10-item SUS questionnaire after the tasks. Completed by 5 testers (T1 to T5); see the sample-size note above. Scored with the standard SUS method: for odd-numbered items, score minus 1; for even-numbered items, 5 minus score; then add the ten results and multiply the sum by 2.5, giving a score from 0 to 100 for each tester. The mean of the testers' scores is reported.
UX3. Accessibility checks (proposed)	Number of critical issues found by an automated accessibility checker (for example axe or Lighthouse) plus a manual keyboard-only run of the main tasks.	0 critical issues (proposed)	Run the automated checker on the prototype's main screens and record the critical issues. A team member then completes the main tasks using only the keyboard and records any task that cannot be finished. Critical issues from both checks are counted together.

HAI principle metrics (User Control and Agency)

Metric	How it is measured	Target	How data is collected
HAI1. Disclosure understood	Share of testers who can say that Masari is an AI, what information it uses, and that the final decision is theirs.	≥ 90% of testers	Three short questions after the session. Answers recorded by anonymous code.
HAI2. Reasons found and understood	Share of testers who find the reasons for a suggestion and explain what influenced it.	≥ 80% of testers without help	Observed task plus one check question.

Metric	How it is measured	Target	How data is collected
HAI3. Time to change a choice	Share of testers who change an answer or a factor's importance and see updated suggestions within 60 seconds.	≥ 80% within 60 seconds	Observer times the task with a stopwatch.
HAI4. Perceived control	Share of testers who rate "I felt in control of the final decision" 4 or 5 out of 5.	≥ 80% rate 4 or 5	One extra item added to the post-test questionnaire.
HAI5. Feedback with a reason	Share of "this does not fit me" (thumbs-down) clicks that include a reason. Depends on a proposed feature.	≥ 50% include a reason Fallback if there are few clicks: at least 3 of 5 thumbs-down clicks include a reason.	Anonymous feedback log from the prototype during user tests.
HAI6. Feedback acted on (proposed)	Share of submitted feedback reasons that the team reviews and records as a change or a decision (for example, a rule or explanation update). Depends on a proposed feature.	≥ 80% of submitted feedback reasons are reviewed and recorded as a change or a decision	Team review of the anonymous feedback log. Each reviewed reason is recorded as a change or a decision.

HAI Principle Ladder v1

Level	How it applies to Masari	Feature	Linked objectives
1. Inform	Masari tells users that it is an AI guide, explains what verified information it uses, clarifies that suggestions come from documented rules, states that it does not predict ability or success, and confirms that final decisions remain with the users and their families.	Start-screen disclosure and "unverified" flags (in Week 2 concept).	HAI1, AQ2
2. Explain	Each suggestion shows the factors that matched, the mismatches, and the uncertainties in plain language. The explanation must come from the rules actually used, not be written afterward.	Reasons shown with each suggestion.	HAI2, AQ1, AQ4
3. Allow control	Users can skip questions, edit answers, adjust the importance of selected factors, compare alternatives, and ask follow-up questions. The suggestions update accordingly.	Editable inputs, factor-importance control, alternatives comparison, “Simple wording” switch.	HAI3, HAI4, AQ3, UX1, UX3
4. Enable feedback	Users can mark a suggestion as not fitting and explain why. The team reviews the reasons to improve the rules and explanations.	Thumbs-down with an optional reason, logged anonymously.	HAI5, HAI6

Tool Choices

The instructor asked which models are used and what the key interface features are. The decisions below are team decisions, drafted with Claude’s help using the Week 2 concept and the project file, and reviewed by the team. The team can change them. Team review of tool choices confirmed: [NEEDS TEAM INPUT]. Claude (Sonnet 5.5) and Gemini (Flash 3.6 Low) were used only to draft the Week 2 concepts, not as prototype models.

Choice	Decision for Plan v1	Reason and limits
Model: hosted API or open-source	**Hosted API: Google Gemini API, using a Flash-class model on the free tier.** The exact model ID must be taken from Google's official documentation. Free-tier status must be confirmed on Google's official pricing page. The exact model ID is then fixed in the README in Week 6. If the quota is cut, switch to Flash-Lite. The model writes guide answers and simple-wording rewrites only. It never ranks pathways.	No hosting or GPU cost, which fits the project's cost rule. Free-tier quotas have changed before, so the model sits behind one adapter and the interface shows a fallback message on errors and time-outs. On the free tier Google may use inputs to improve its products. Masari never asks for personal data, tests use fictional or anonymous input, and the interface says so.
Backend	**Python with FastAPI**, in three parts: 1. Scoring engine: deterministic rules over the pathway data. 2. Guide endpoint: retrieval, model call, and answer checks. 3. Logging: anonymous feedback and test logs. The API key is kept in an environment variable, not in the repository.	Python is listed in the project file and makes the evaluation scripts easy. Keeping scoring apart from the model makes AQ3 testable.
User interface	**React (Vite) single-page app** with accessible chat markup, right-to-left layout, full keyboard use, messages announced to screen readers, and text that grows to 200%.	Editing answers and weights changes many parts of the screen at once, which suits React state. The Gemini draft also proposed React.

Choice	Decision for Plan v1	Reason and limits
Grounding	**Two layers, both from the same verified source.** (a) Structured pathway data (JSON) written from Decrees 150/2026 and 213/2025 after comparison with the Official Gazette. Each record stores the article, source link, and date checked. It feeds the scoring and the pathway cards. (b) Guide answers use keyword search (BM25) over a few dozen short passages from the same decrees, each with a source ID. The model sees only the retrieved passages and must cite their IDs. If nothing is relevant enough, Masari says it is not sure and points to the school counselor or the Ministry. BM25 on Arabic text needs normalization (for example, removing diacritics and unifying alef and hamza forms), and this is tested in the 30-prompt set.	The text collection is small, so a vector store is not needed. Facts that are not yet verified are flagged, as decided in Week 2.
Safeguards	A rule-based check catches requests to decide for the student or predict ability, admission, or success. The system prompt repeats this. An output check rejects answers that cite a source ID that does not exist.	Supports AQ2 and AQ1.
Language	**Arabic by default (right-to-left), with an English option.** Pathway names appear in Arabic with an English translation.	Matches the personas' needs. Arabic answer quality is checked in the test set.

Interface key features (Week 2 selected concept)

Feature	How it supports User Control and Agency
System overview	Plain explanations of the system, with unverified details flagged. Students move through it at their own pace.
Pathway cards and comparison	Same criteria for every pathway. No pathway is labeled better.
Interactive assessment	Short, non-judgmental questions. Students can skip questions and revise answers.

Feature	How it supports User Control and Agency
Explainable suggestions	A small number of pathways to explore, with matching factors, mismatches, and uncertainties. Students adjust the importance of selected factors.
Constrained conversational guide	Follow-up questions, comparison of alternatives, and answers limited to verified information.
Notes	Students record their own thoughts for later discussion. The final decision stays with the student.

Week 4 Prompt Log

Field	Details
AI tool name	Claude (Sonnet 5.5)
Exact prompt	The exact prompt and attached files are recorded in Entry W3-1 in Appendix A of the Week 3 document and are not repeated here. According to that record, one request produced both the Week 3 and Week 4 deliverables. The tool-choices request is logged separately in Entry W4-2, and the revision request for this document in Entry W4-3.
What it gave (actual output)	The Week 4 parts of the same Word document: journey maps, measurable objectives with proposed targets, HAI Principle Ladder v1, and tool choices.
What we changed and why	**The document was updated to reflect the revised personas and the changes in the project's target users.** 1. **Personas: Replaced Mariam and Yara with Hoda Ramadan (parent) and Hassan Nour (school counselor), while keeping Omar (student). This change ensures that the personas represent the three primary user groups: students, parents, and school counselors.** 2. **Journey Maps: Updated the Week 4 Journey Maps to represent Omar, Hoda, and Hassan instead of Omar, Mariam, and Yara. This ensures that each journey reflects the goals, needs, challenges, and interactions of the current personas.** 3. **User Roles: Revised the wording to identify parents and school counselors as primary users rather than secondary supporters, since they have their own goals and interact directly with Masari.** 4. **User Testing: Updated the testing plan so that adult testers from outside the team role-play the three primary**

Field	Details
	user groups: a Grade 9 student, a parent, and a school counselor. Clarified that no minors are involved and that the results may not fully represent the experiences of real users. **5. HAI Ladder and Metrics: Replaced student-specific wording with user-focused language where appropriate, ensuring that the Inform, Explain, Allow Control, and Enable Feedback levels apply to all three primary user groups.** **These changes were made to improve consistency across the document and ensure that its personas, user roles, design decisions, and evaluation plans reflect the current scope of Masari.**